Replacing the Linear Time Term in the Analysis of the Hybrid N-of-1 Design: Advantages and Disadvantages of a Flexible Time Adjustment

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Purpose. The published analysis of Hendrickson et al. [1] adjusts for symptom change unrelated to the drug with one linear term in time, `t`. This paper weighs replacing it with a more flexible adjustment: a spline in time, visit as a factor, or terms tied to the trial's expectancy phases. It draws on the Monte Carlo results of `docs/40`, Section 9 (script `07-time-adjustment.R`: 27,000 fits on alternative cells and 12,000 on null cells, all converged) and on the population-mean calculations of the same paper.

(Updated 2026-10-09: results at $t_{1/2} > 0$ now use anchored carryover; see `docs/53`. Values changed by at most 0.008 in rejection rates and 0.53 points in the drug coefficient β_D (the interaction estimates by at most 0.0003); conclusions unchanged.)

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md` and `docs/40`. Symbols marked *local* are defined here. (Notation, 2026-10-09: the linear-time slope β_t is now β_T, as in `docs/56` and the compendium.)

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score (code `Sx`)	canonical
TV, PB, BR	natural-course, expectancy and drug responses	canonical
w_t	calendar week of visit t (code `t`)	canonical
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `e`, regressor `De`); written e_t before 2026-10-07	canonical
D_{it}	binary drug state (code `Db`)	canonical
β_D, $\beta_{bm:D}$	drug main effect; biomarker-by-drug interaction	canonical
$f(t)$	the time adjustment: $\beta_T w_t$, a spline, or visit effects μ_t	local
β_T, μ_t, β_η	linear-time slope (written β_t before 2026-10-09); visit effect; expectancy coefficient	canonical
$t_{1/2}$	carryover half-life (weeks)	canonical
N	total sample size across paths	canonical
ρ	within-component AR(1) correlation per week (configuration C)	canonical

- **Linear time.** $f(t) = \beta_T w_t$, the published adjustment.
- **Spline.** A natural cubic spline in w_t, here with 3 degrees of freedom (`ns(t, df = 3)`).
- **Visit as factor.** One free mean per visit, μ_t (`factor(t)`).
- **Expectancy term.** $\beta_\eta \eta_t$ (`De`), present in the published code but switched off for the publication.
- **Size.** The rejection rate when there is no interaction (Type I error); nominally 0.05.

1. Summary

1. **The linear term stands in for two curves and a step.** The non-drug mean is a sigmoid natural-course curve plus an expectancy curve that halves at week 9, when blinding begins. A straight line leaves a root-mean-square misfit of 3.2 points in the population mean of the Hybrid design. A spline leaves 1.9 and visit effects 0.8 (Section 2).

2. **For the interaction, flexibility buys nothing in bias.** In every cell the three adjustments give the same mean interaction estimate, within 0.0009, because the biomarker is unrelated to the non-drug components (Section 4).
3. **For the drug main effect, flexibility removes a bias of about 2 points.** Linear time credits the open-label expectancy to the drug. Visit effects replace this with a randomized estimate from the between-path comparisons, at about 1.5 times the standard deviation (Section 4).
4. **In Hybrid, the flexible adjustments are anticonservative under the published random-intercept covariance.** Under one-week carryover the Type I error is 0.075 (spline) and 0.080 (visit factor), against 0.056 for linear time. Their higher power, 0.02 to 0.07, is partly this inflation. In the crossover all three hold their size, and the flexible forms gain 0.01 to 0.04 (Section 4).
5. **The flexible adjustment and the working covariance have to be chosen together.** A random intercept misrepresents the outcome covariance. The flexible adjustments remove the mean misfit that, on our working explanation, had been masking that error. Visit effects paired with an unstructured covariance, the MMRM convention, are the natural form, but their size in this design has not yet been tested (Sections 5 and 7).
6. **Recommendation.** For the interaction test, keep linear time under the published covariance, or move to visit effects only together with an unstructured covariance and a size check. For the drug main effect, use visit effects or add the expectancy term (Section 8).

Update (2026-10-07). The size check of point 5 has been made (`docs/45`, Section 4.4). In the same cell (CS construct, Hybrid, $t_{1/2} = 1$, 500 null replicates, MCSE 0.0097), the excess recurs under model-based inference (`cs-Satt`: 0.054 linear, 0.070 spline, 0.072 visit factor) and under the unstructured covariance with Kenward-Roger (0.056, 0.068, 0.072), is smaller with baseline as a covariate (0.056, 0.060, 0.064), and is absent under CR2 (`cs-CR2` 0.058 and 0.058; `csh-CR2` 0.060 and 0.060, linear and visit factor). Seen in two separate simulation runs with the same direction and size, the excess is real for model-based and Kenward-Roger inference. `docs/45` calls this evidence "weak" because the eight-cell mean size is unchanged; that reading is corrected here, since an excess confined to the carryover cell is still an excess. The unstructured covariance therefore does not license visit effects, and recommendation 2 of Section 8 is withdrawn: a flexible time adjustment should be paired with CR2 inference. The project's current primary analysis, F1 of `docs/46` and `docs/49`, uses linear time with CR2 inference and adds the expectancy term `De` with its biomarker interaction.

2. What the time term has to absorb

The published model is `Sx ~ bm + Db + t + bm*Db + (1|ptID)` (`docs/40`, Section 8). Its time term must absorb the part of the mean that does not depend on the drug state. In the Hybrid design that part is the same in every path, because all paths share the visit schedule and the blinding pattern (`docs/40`, Section 3):

Table 2: Population means of TV, PB and their sum by week, common to all paths

Week	4	8	9	10	11	12	16	20
TV	1.86	4.79	5.24	5.59	5.85	6.03	6.39	6.48
PB	1.86	4.79	2.62	2.79	2.92	3.02	3.19	3.24
$TV + PB$	3.73	9.58	7.86	8.38	8.77	9.05	9.58	9.72

(Corrected 2026-10-08: the week-4 sum was given as 3.72, the sum of the rounded components; the exact value is 3.73.)

The sum rises steeply to week 8, falls by 1.7 points at week 9 when expectancy halves, and then flattens. No straight line follows that.

A time adjustment also absorbs the part of the drug response that one `Db` coefficient cannot represent at visits where every path is in the same drug state. At weeks 4, 8 and 9 every path is on drug, with a drug response that rises from 4.3 to 9.8 on the Gompertz curve. A flexible adjustment takes up that rise; linear time cannot. This is why the choice changes the drug coefficient so much (Section 4).

Root-mean-square misfit of the population mean (allocation-weighted least squares with `Db` in the model, no carryover; verified, `docs/40`, Figure 7):

Table 3: Misfit of each time adjustment to the population means, Hybrid design, no carryover

Time adjustment	Linear	Linear + `De`	Spline (3 df)	Visit factor
Misfit (points)	3.18	3.11	1.93	0.80

What visit effects still miss is the drug response itself, which varies from 4.3 to 10.2 across visits while `Db` is one number.

3. The options

Table 4: Time adjustment options with parameter counts, what each absorbs, and identifying contrasts

Adjustment	Parameters (Hybrid)	Absorbs	Identifies β_D from
Linear `t`	1	the overall slope	all on-off contrasts, within and between paths
Linear `t` + `De`	2	slope and the expectancy step	as above
Spline, `ns(t, 3)`	3	smooth curves, not the step	as above
Visit factor	8	everything common to all paths at a visit	between-path contrasts at weeks 10, 16, 20 only
Phase indicators	2 or more	level shifts between open label, blinded discontinuation and crossover	within-phase contrasts
Random slope on `t`	2 variance terms	participant-specific linear trends	as for linear `t`

The published code already offers two of these: `useDE = TRUE` adds the expectancy term, and `t_random_slope = TRUE` adds a random slope on time. Neither was used for the publication (`58b32a9` vignette 1, line 350).

4. Evidence

All results are for the published compound-symmetry construct with $N = 70$ and the published allocation, fitted with `lmer` and a random intercept. Interaction strength is 0.25 in the alternative cells, with 500 replicates per alternative cell and 1000 per null cell (`docs/40`, Section 9).

The interaction estimate does not depend on the adjustment. The three mean estimates of $\beta_{bm:D}$ agree to within 0.0009 in every cell. In Hybrid their empirical SDs agree too (graded coupling, $t_{1/2} = 0$: 0.0562, 0.0559, 0.0558). The biomarker is unrelated to TV and PB, so misfit of the time term adds noise to the on-off contrast but no bias, and it does not even add measurable noise to the estimate.

The drug effect depends heavily on it (Hybrid, graded coupling):

Table 5: Drug effect limits and simulated means by time adjustment, Hybrid design, graded coupling

Adjustment	Limit, $t_{1/2} = 0$	Simulated mean (SD)	Limit, $t_{1/2} = 1$	Simulated mean (SD)
Linear `t`	−8.21	−8.16 (0.86)	−6.38	−6.31 (0.84)
Linear `t` + `De`	−7.28	not run	−4.81	not run
Spline	−6.22	−6.02 (0.99)	−3.67	−3.50 (0.94)
Visit factor	−6.25	−5.99 (1.27)	−4.49	−4.28 (1.19)

'Limit' is the value each model converges to, computed from the population means. In the crossover, where expectancy is constant, the three adjustments agree within 0.5 points.

Power and size of the interaction test:

Table 6: Power and size of the interaction test by design, cell and time adjustment

Design	Cell	Linear `t`	Spline	Visit factor
Hybrid	null, $t_{1/2} = 0$	0.049	0.055	0.059
Hybrid	null, $t_{1/2} = 1$	0.056	0.075	0.080
Hybrid	graded, $t_{1/2} = 0$	0.636	0.678	0.690
Hybrid	graded, $t_{1/2} = 1$	0.442	0.488	0.510
Hybrid	mean moderation, $t_{1/2} = 2$	0.626	0.674	0.690
CO	null, $t_{1/2} = 0$	0.052	0.040	0.044
CO	null, $t_{1/2} = 1$	0.042	0.040	0.039
CO	graded, $t_{1/2} = 0$	0.734	0.750	0.772

Monte Carlo SEs are about 0.007 for the null rates and 0.021 for power. The power, size and drug-effect figures of this section were checked against `analysis/data/quick-sim/carryover-closed-form/time-adjustment/` (`summary-reps500.csv`, `summary-reps1000-null.csv`; verified 2026-10-07).

5. Advantages of a flexible time adjustment

1. **It removes the confounding of the drug effect with expectancy.** Linear time overstates the drug effect by about 2 points in Hybrid, because the open-label visits are both on drug and at full expectancy, and a line cannot follow the drop at week 9. Visit effects remove the confounding entirely. Adding `De` removes half to four fifths of it (Section 4; "two thirds" before the 2026-10-09 anchored-carryover update, as in `docs/40`).

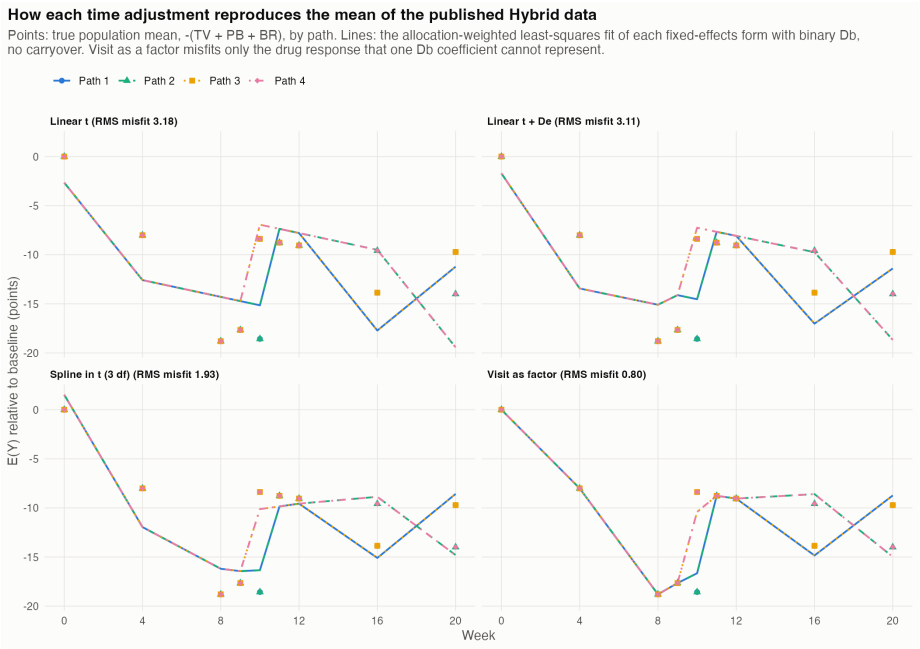


Figure 1: True population means by path (points) and the best fit of each time adjustment (lines), Hybrid, no carryover (reproduced from docs/40, Figure 7).

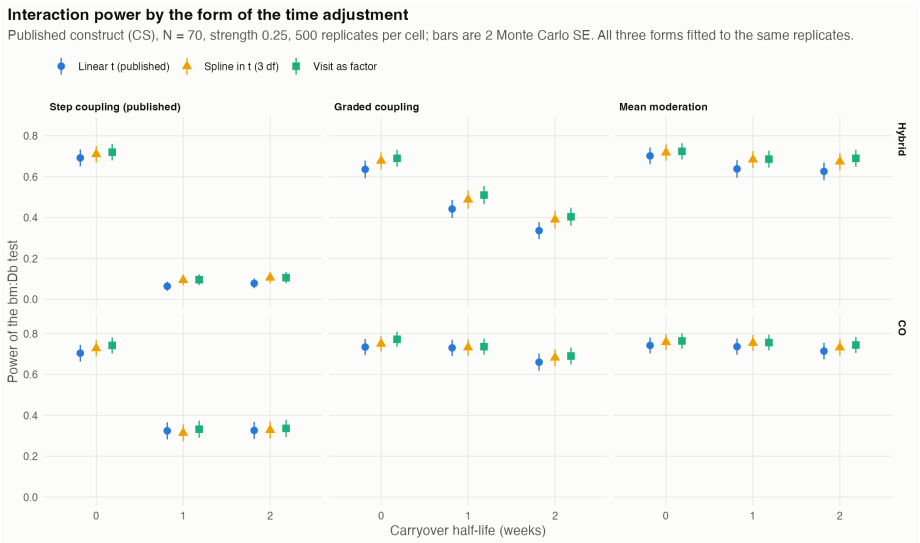


Figure 2: Interaction power by time adjustment, with 2-SE bars (reproduced from docs/40, Figure 8).

2. **It gives a randomized drug estimate.** With visit effects, the drug effect rests on comparisons between paths at the same visit (weeks 10, 16, 20). Those comparisons are protected by the randomization of paths rather than by a model of the time trend. This also makes the analysis robust to the trend's shape, which is unknown in a real trial (`docs/31`, Section 4, on why natural history rarely follows a line).
3. **It matches the MMRM convention.** Visit as a factor with an unstructured covariance is the standard analysis of a longitudinal trial with fixed visits, and period effects are a first-order concern in within-participant designs [3]. Reviewers recognize it, and it can be specified without choosing a functional form.
4. **It improves the crossover analysis.** In CO the flexible adjustments hold their size and gain 0.01 to 0.04 in power, partly from a genuine reduction in the estimate's variability (SD 0.0487 against 0.0462 for graded coupling at $t_{1/2} = 0$).
5. **It costs few parameters.** Seven extra parameters for visit effects against 630 observations at $N = 70$, or two for a 3-df spline.

6. Disadvantages of a flexible time adjustment

1. **Under the published covariance it inflates the Type I error in Hybrid.** Spline and visit effects reject 7.5% and 8.0% of null trials under one-week carryover, about 3.6 and 4.3 Monte Carlo SEs above nominal (MCSE 0.0069 at a true rate of 0.05; the earlier "3 and 3.5" used the SE at the observed rates), against 5.6% for linear time (Section 4 table; updated 2026-10-09 to the anchored-carryover rerun, previously 7.3%, 8.0% and 5.8%). Without carryover the excess is within about one SE. The excess was later reproduced, and shown to vanish under CR2 (see the update after Section 1). The cause is not established. On our working explanation (inferred, not tested), the random intercept misrepresents the outcome covariance: the baseline row has no noise of its own, and the variance falls at blinding (`docs/40`, Section 6). The mean misfit left by linear time inflates its residual variance enough to offset this, and the flexible adjustments remove that offset.
2. **Its power gains in Hybrid are not clean.** The estimates are no less variable than linear time's, yet they reject more often. Their model-based SEs are therefore smaller, consistent with the size inflation. No size-adjusted comparison has been made.
3. **Visit effects weaken the drug estimate.** Identified only by the between-path comparisons, the drug effect's SD rises from 0.86 to 1.27 in Hybrid. With a single path the drug state and the visit effects are completely aliased and the drug effect cannot be estimated. With fixed allocation, every path must be represented.
4. **A spline cannot follow the expectancy step.** A smooth curve through the week-9 drop absorbs part of it into the drug effect. Under

carryover the spline's drug effect falls below the visit-factor value (-3.67 against -4.49 in the limit at $t_{1/2} = 1$). Splines also need a choice of degrees of freedom or knots, which should be fixed rather than tuned to the data.

5. **It does not help the interaction estimate.** In this construct the interaction is already unbiased under linear time. A flexible adjustment protects the interaction only if the biomarker is related to the time trend, for example a severity biomarker that predicts regression to the mean. Then any misfit of the trend loads onto $\beta_{bm:D}$. The construct has no such link (`docs/31`, Section 18.1), so the simulations cannot show that benefit.
6. **Visit effects need fixed visits.** In a real trial with visit windows, timing varies around the nominal week. Visit effects then absorb the nominal visit, and the within-window timing is ignored. A spline in actual time handles this better.

7. Interplay with the covariance and the baseline coding

The flexible time adjustment cannot be judged on its own. Three choices interact:

- **Covariance.** The random intercept implies compound symmetry, which the outcome does not have, even under the compound-symmetry construct (`docs/40`, Section 6). An unstructured covariance (`us` in `mmrm`) represents any phase structure. The anticonservatism of Section 6 may disappear under it; that is the main open question. *Answered (2026-10-07):* it does not; with Kenward-Roger the excess persists (`docs/45`, Section 4.4; see the update after Section 1).
- **Baseline coding.** The baseline row is the worst-fitting row of the covariance and has the most leverage on a linear trend. With visit effects, baseline has its own mean and does not inform the drug contrast. Baseline as a covariate (`docs/41`) removes the row altogether.
- **Expectancy term.** `De` is a one-parameter step that captures most of what linear time misses in the drug effect. It has a moderate correlation with `Db` (0.59) and a variance inflation of 1.5 (`docs/40`, Section 8). The paper's concern about collinearity [1] does not rule it out at $N = 70$.

The coherent combination is the MMRM: visit effects, an unstructured covariance, and baseline as a covariate or as a cLDA row. Each part removes one misspecification, and none is known to cause the size problem of Section 6. Whether the combination holds its size in the Hybrid design is what the paused covariance study (`08-covariance-study.R`) is designed to show. *Update (2026-10-07):* that study is complete (`docs/45`). With visit effects, the unstructured covariance and Kenward-Roger reject 0.072 in the Hybrid carryover cell, and 0.064 with baseline as a covariate; the combination does not hold its size there.

8. Recommendation and open questions

For the interaction test:

1. Under the published random-intercept covariance, keep linear time. It holds its size in this construct (0.049 and 0.056), and the flexible forms' extra power is partly inflation.
2. Move to visit effects only together with an unstructured covariance, and verify the size by simulation first. With 70 participants and 45 covariance parameters, use small-sample degrees of freedom (Kenward-Roger [2] or Satterthwaite) for the test. *Withdrawn (2026-10-07):* the verification failed (0.072 with Kenward-Roger in the Hybrid carryover cell, `docs/45`, Section 4.4). If a flexible time adjustment is required, pair it with CR2 inference about a parsimonious working covariance, which was nominal in that cell. The unstructured model with Kenward-Roger also failed the stress test of `docs/46` with linear time (pooled 0.057; 0.087 at $N = 35$).

For the drug main effect:

3. Use visit effects, or at least add `De`. Linear time alone overstates the drug effect by about 2 points in Hybrid. Report which estimand is used, since the drug effect varies across visits and each adjustment weights it differently.

Open questions:

- Does an unstructured covariance remove the anticonservatism of visit effects in Hybrid, and does baseline as a covariate help further? *Answered (2026-10-07; `docs/45`, Section 4.4):* no, and only partly (0.064); CR2 removes it.
- How do the adjustments compare under an AR(1) construct (configuration C, at $\rho = 0.8$ in `docs/36`), where the published analysis is itself anticonservative without a serial-correlation term (`docs/36`, Section 6.8)?
- What is the power of each adjustment at matched size?
- Does a random slope on time (`t_random_slope = TRUE`) behave like a flexible fixed adjustment? It was not tested.

9. Limitations and evidence status

- **One construct.** All simulations use the published compound-symmetry construct with strength 0.25 (verified). Under AR(1) data the size results may differ.

- **One covariance in the evidence.** All fits in Section 4 use a random intercept. The comparison with unstructured covariance is the subject of the paused study and is not reported here (since reported in `docs/45`; see the update after Section 1).
- **The mechanism of the anticonservatism is inferred.** It has not been tested by refitting with a heteroscedastic or unstructured covariance. The later refits (`docs/45`) show that `cs` with model-based inference and `us` with Kenward-Roger do not remove it and CR2 does, which argues against a purely covariance-structure explanation; the mechanism remains untested.
- **No size-adjusted power** in this document. `docs/45` (Section 5) reports size-adjusted power for linear time only.
- **Phase indicators and random slopes are described, not simulated.** Paper 06's phase-augmented model, with phase-by-drug interactions, is a different question (attribution, `docs/31`, Section 11) and is not evidence about the time adjustment.

10. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.
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